

Proposed Solution Report (Today's Work): Cloud-Based Movie Streaming Platform Using FastAPI and Next.js

Mustakim Ahmed Hasan (2421349042), MD. Nafiz Al Mamun (2232050642),
Tanim Ahmed (2312597042), Ahamed Osman Asif (2031080642)
CSE299.4; Group: 01
Department of Computer Science and Engineering
North South University

I. PROBLEM STATEMENT

Two issues were impacting user experience and deployment stability:

- The navbar briefly showed login/signup even for authenticated users after reload.
- Search only returned exact text matches instead of relevant or typo-tolerant results.
- Frontend build failed due to server cookie handling incompatibilities.

II. PROPOSED SOLUTION

We propose targeted fixes focused on authentication state consistency, search relevance, and build stability:

- Seed auth state on first render using server-side cookies and bearer token support.
- Add fuzzy, ranked search on the backend to return relevant results even for typos.
- Correct server cookie handling to pass production builds consistently.

III. SOLUTION DETAILS

A. Auth State Consistency

- Use server-rendered auth check to avoid flicker on reload.
- Store access token on the frontend domain and send it as Authorization header.
- Make backend accept both cookies and bearer tokens.

B. Search Relevance

- Normalize titles/descriptions and compute similarity scores.
- Rank by exact match, prefix match, token overlap, and fuzzy similarity.
- Preserve pagination and frontend API contract.

C. Build Stability

- Fix Next.js cookie access by awaiting cookies() and using safe fallbacks.
- Validate production build locally.

IV. EXPECTED OUTCOME

- Navbar no longer shows incorrect login/signup state after reload.
- Search returns relevant results even for typos or partial queries.
- Frontend builds cleanly for production deployment.